SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1110

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture. *(BL3)*

Assessment Tool Weightage: 60%

QUESTIONS: 6 **Total Marks:60**

Annexure A

Application Based Questions

Code of Conduct:

I Tejaswini.M(192411079) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Tejaswini.M

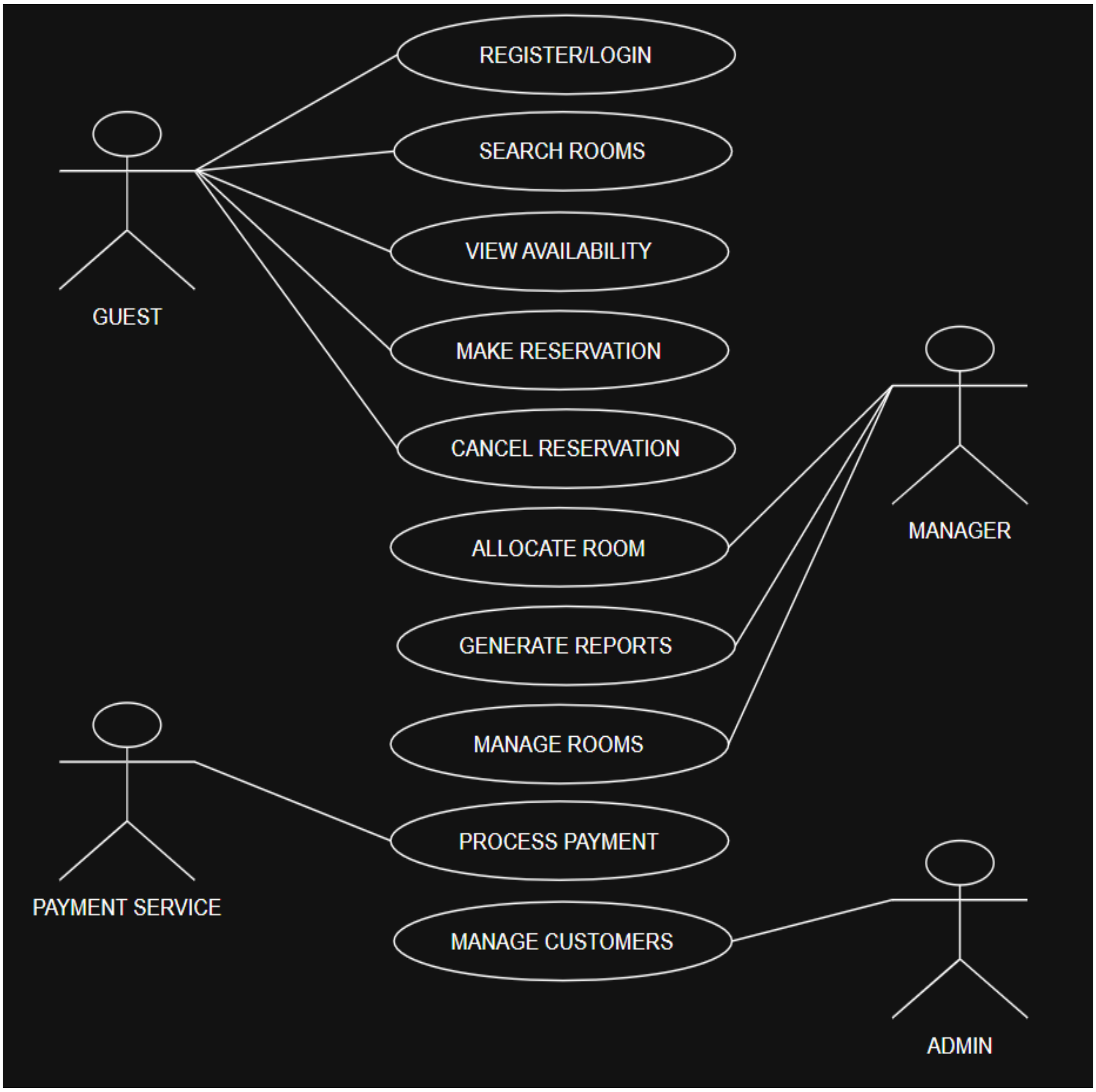
Annexure A

Application Based Questions

1. Use Case Diagram Analysis

A **Smart Hotel Reservation System** allows guests to search rooms, make reservations, cancel bookings, and process payments. Managers handle room allocation and reports, while payment services process transactions. The current design does not clearly distinguish actor responsibilities.

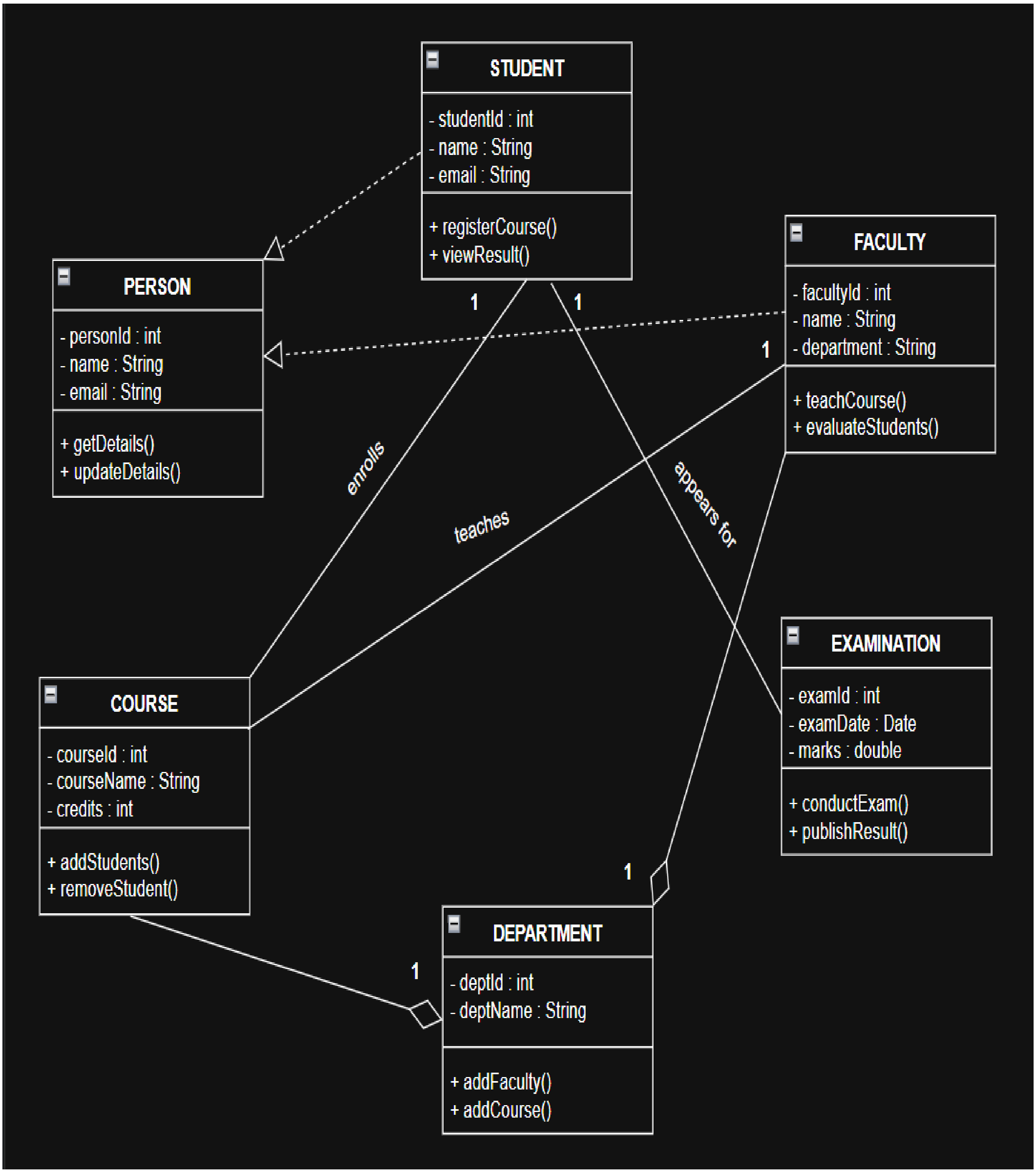
Analyze the system requirements, identify missing actors and use cases, and design an optimized Use Case Diagram showing proper relationships (include, extend, generalization) for better system clarity.



2. Class Diagram Design

A **University Management System** includes entities such as Student, Faculty, Course, Department, and Examination. The existing design lacks proper attributes, operations, and relationships among classes.

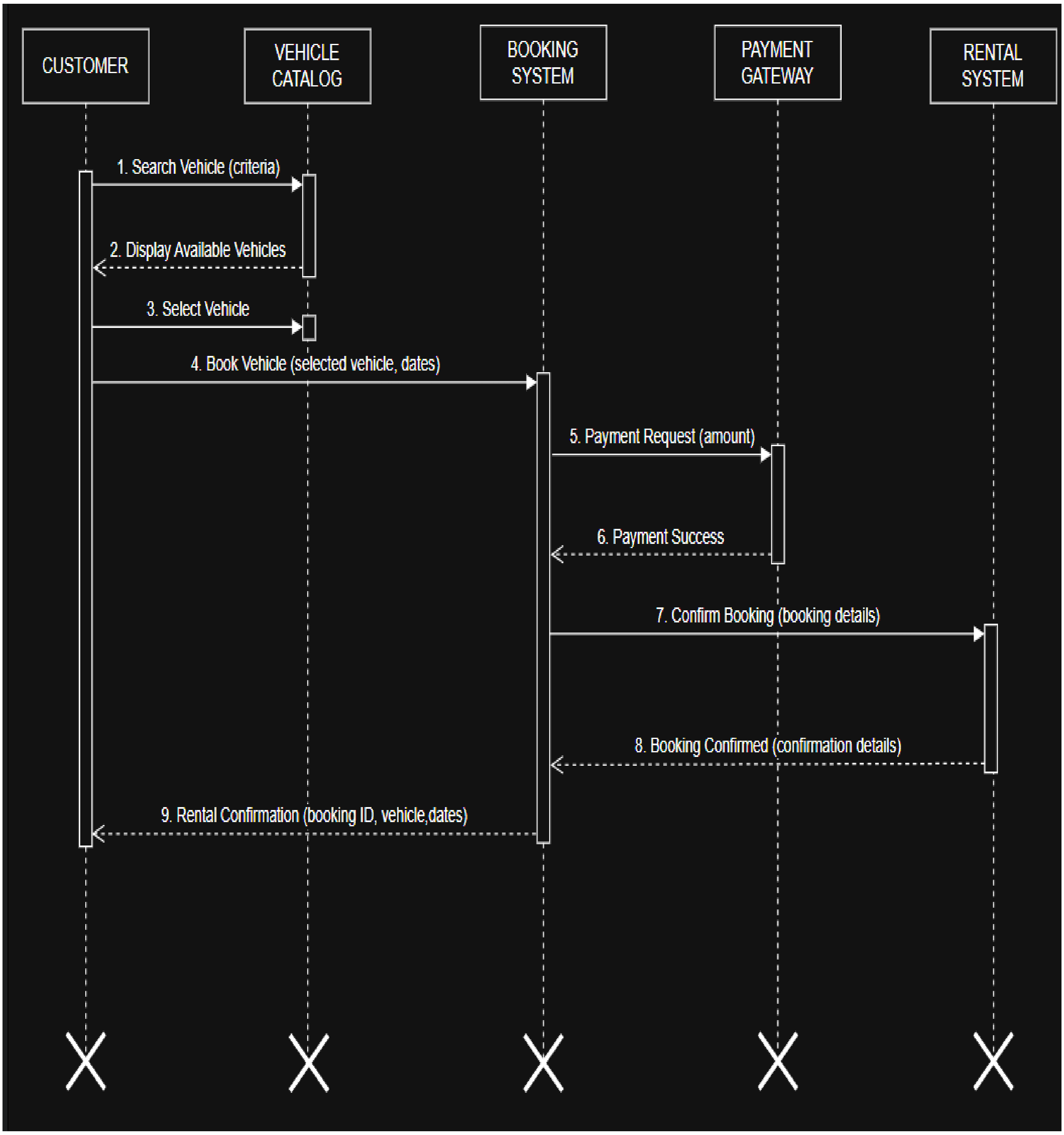
Analyze the system, identify relevant classes with attributes and methods, define relationships (association, aggregation, inheritance), and construct a well-structured Class Diagram.



3. Sequence Diagram Evaluation

In a **Vehicle Rental System**, a customer searches for a vehicle, books it, makes payment, and receives a rental confirmation. The interaction among system components is incomplete and poorly documented.

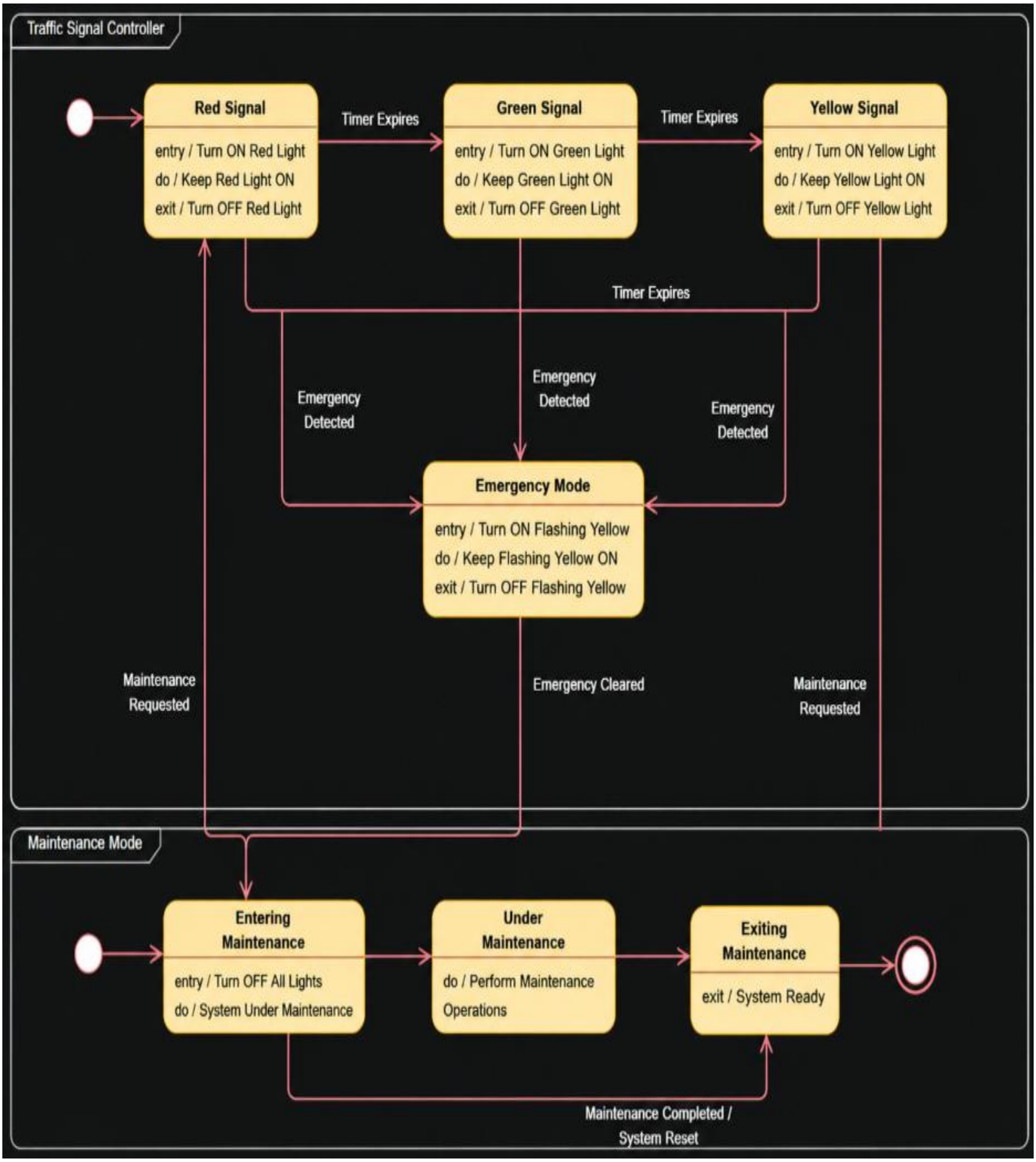
Analyze the process flow, identify missing objects and message sequences, and design a detailed Sequence Diagram representing proper interaction among system components.



4. State Diagram Construction

A **Traffic Signal Control System** operates through states such as Red Signal, Green Signal, Yellow Signal, Emergency Mode, and Maintenance Mode. The transitions between states are not clearly defined.

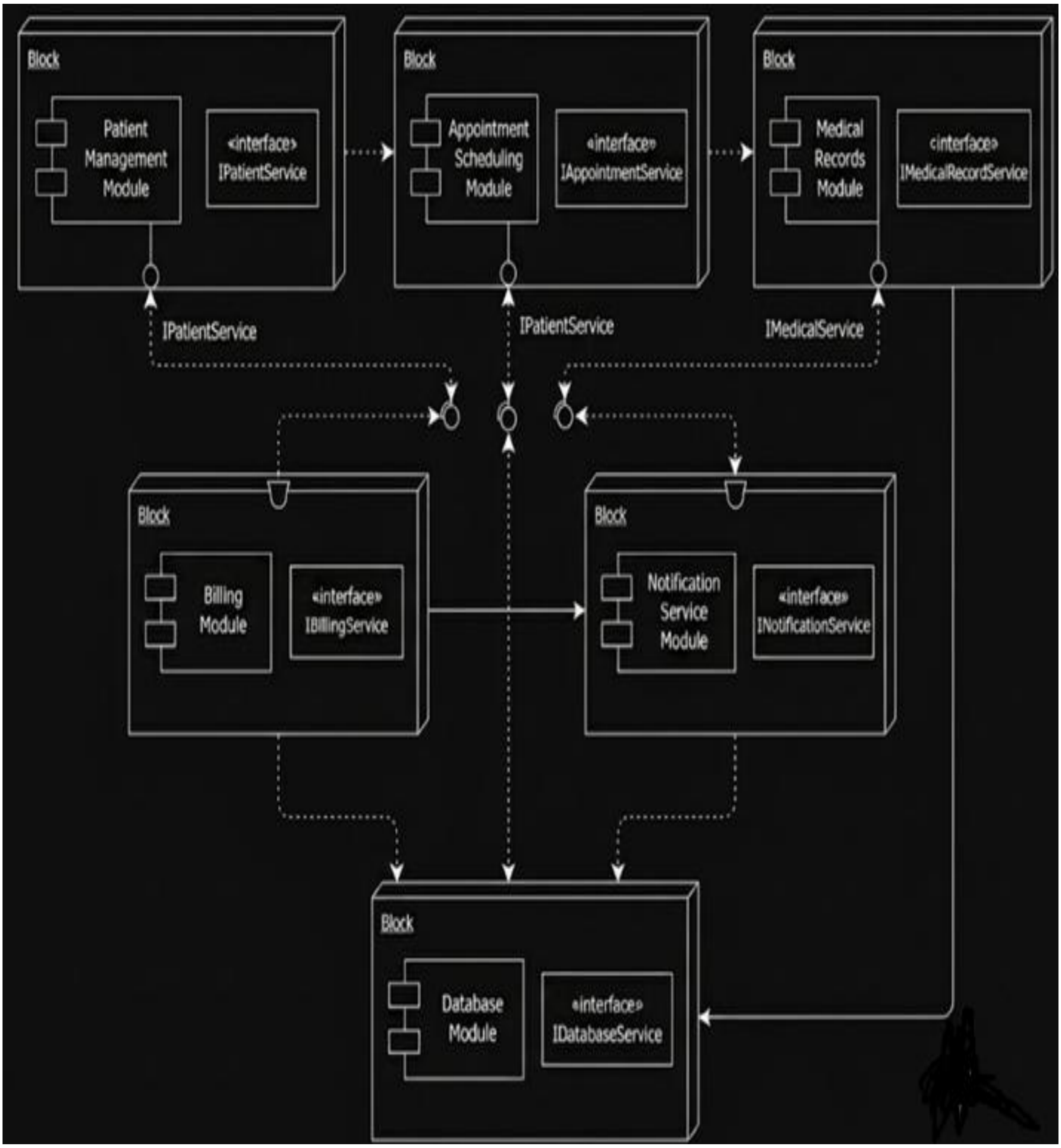
Analyze the system behavior, identify valid states and transitions, and construct a State Transition Diagram ensuring all events and conditions are properly represented.



5. Component Diagram Design

A **Smart Healthcare Portal** consists of modules such as Patient Management, Appointment Scheduling, Medical Records, Billing, and Notification Services. The current architecture does not clearly define component interactions and dependencies.

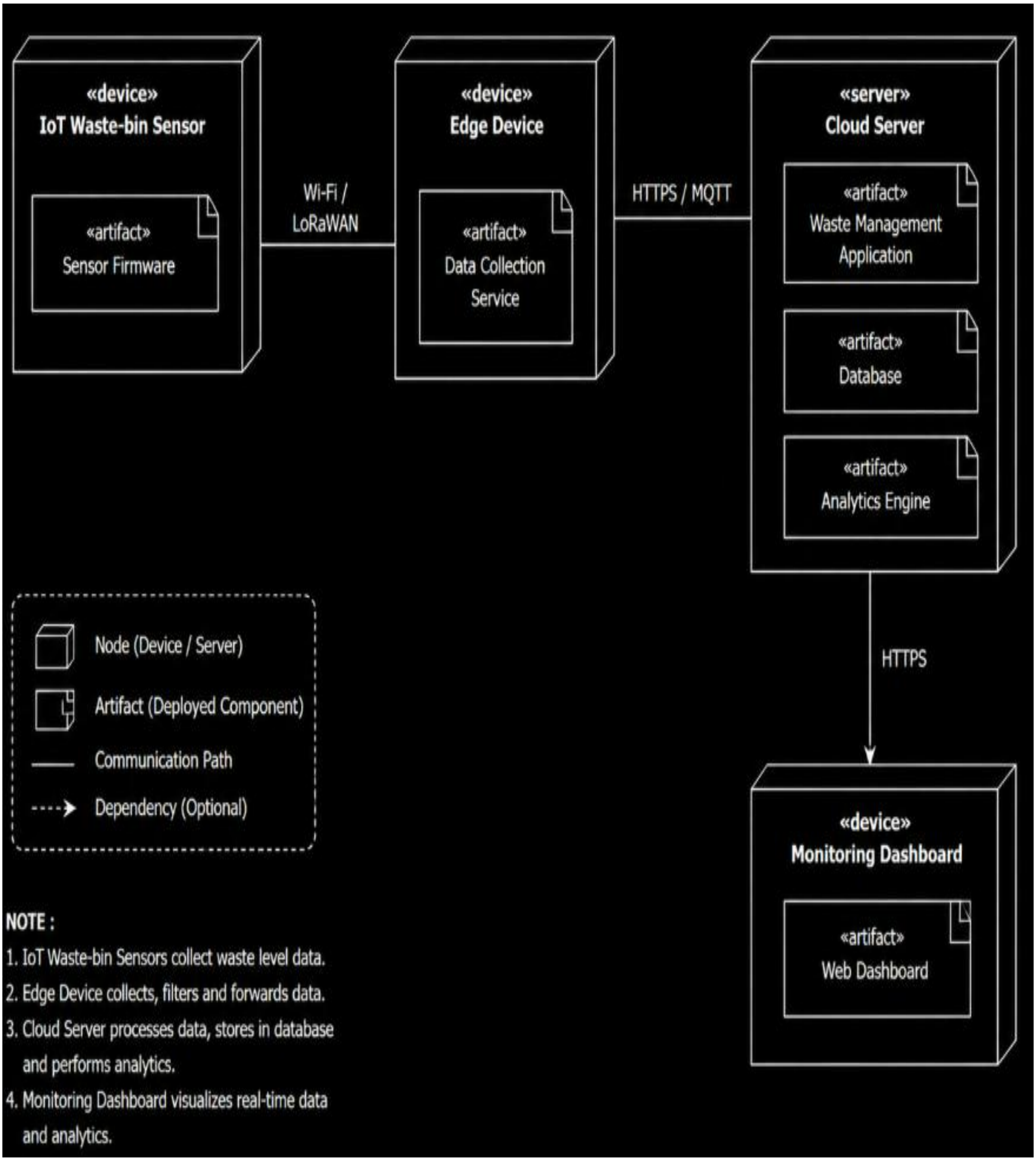
Analyze the system architecture, identify major components and their interactions, and design a Component Diagram showing dependencies and interfaces between modules.



6. Deployment Diagram Design

A **Smart City Waste Management System** is deployed across IoT waste-bin sensors, edge devices, cloud servers, and monitoring dashboards. The current design lacks clarity regarding physical architecture and communication links.

Analyze the deployment environment, identify nodes, artifacts, and communication links, and construct a detailed Deployment Diagram representing the complete system architecture.



RUBRICS FOR CALCULATION

Application Based Questions

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand